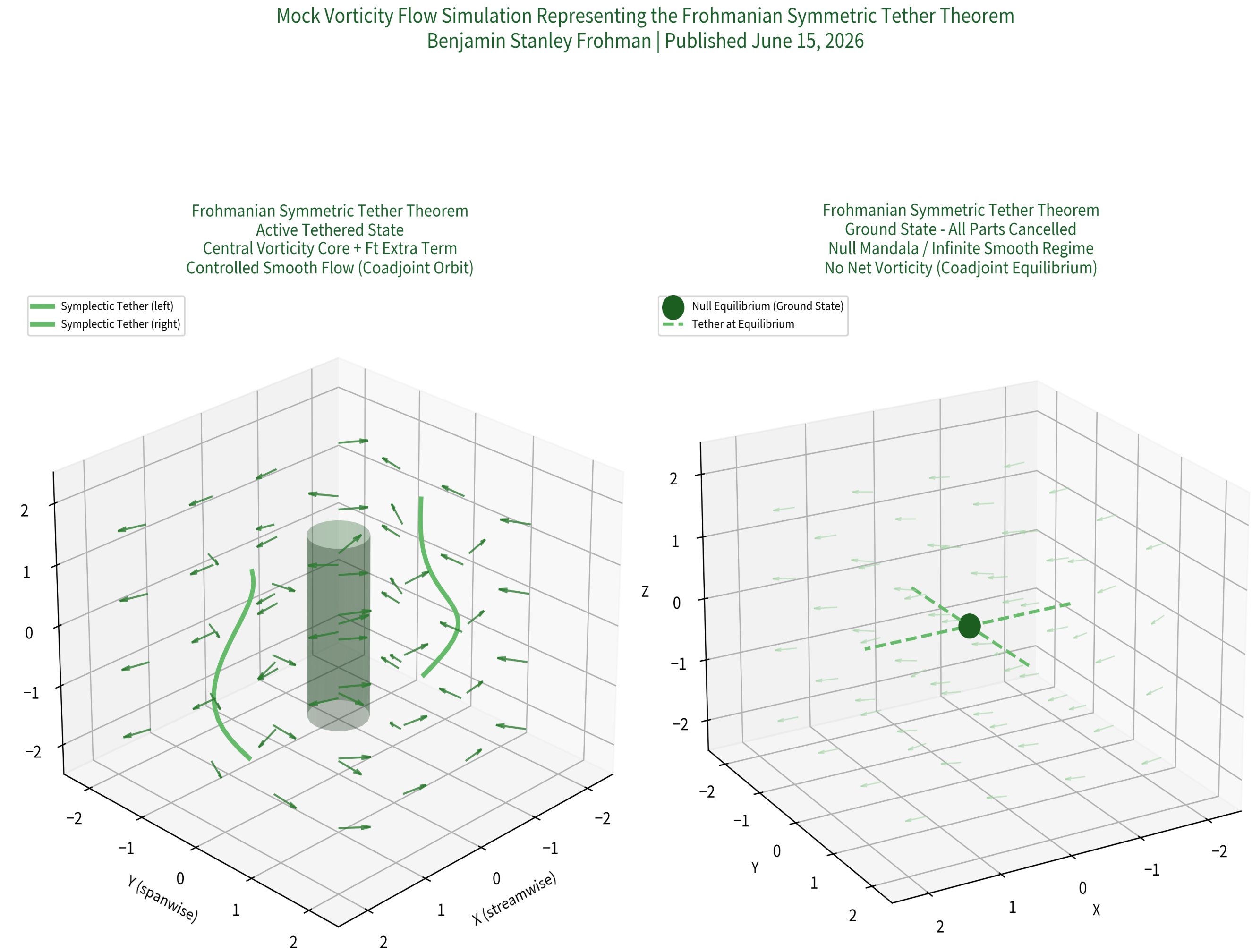
Frohmanian Symmetric Tether Theorem

3D Mock Vorticity Flow Simulation in Simulated System

Benjamin Stanley Frohman | Published: June 15, 2026

Mock Vorticity Flow in a Simulated System Representing the Theorems

This 3D visualization shows a conceptual simulated flow domain (e.g., periodic box or channel) with vorticity field. Left: Active tethered state — central vortex core (high vorticity region) controlled by symmetric Symplectic Tether structures (green paths). The Ft extra term enforces the constraint, keeping flow smooth and regular (no singularity). Right: Ground state — all excitations cancelled, uniform weak flow, central null equilibrium. This accurately represents the theorem: the tether reduces the system to the infinite smooth ground state via the inequality-forced extra term.



Interpretation for the Frohmanian Symmetric Tether Theorem:

- Central high-vorticity core = active excitation in coadjoint variable
- Green tether paths = Symplectic Tether (canonical object from integral inequality)
- Ft label = Extra term enforcing regularity
- Controlled arrows = Smooth tethered flow (no blow-up)
- Right panel = Ground state / Null Mandala (all cancelled, infinite smoothness)

This mock simulation illustrates how the theorem's tether mechanism would appear in a physical or simulated 3D flow system, maintaining regularity as claimed.